



Caelus

LIVING WEATHER

Caelus User Guide

Live station screenshots · September 9, 2026

Caelus is a local weather dashboard and history service for an Ecowitt-compatible gateway. It combines current station readings, forecasts, 24-hour metric cards, longer-range graphs, a regional Windy radar map, and observer-local Sun and Moon information in one browser view.

This guide is for day-to-day users. You do not need to understand Python, FastAPI, or SQLite to operate Caelus.

The screenshots were captured from a running Caelus station on September 9, 2026. Your location, readings, gateway model, forecast, lunar view, and available metrics will differ.

Open Caelus

Start Caelus with the launcher created by the installer, then open one of these addresses:

- On the Caelus computer: `http://127.0.0.1:8767`
- From another device on the same trusted network: `http://<Caelus-computer-LAN-IP>:8767`

The usual installed launchers are `/Users/<name>/Caelus/run_caelus.sh` on macOS, `/home/<name>/Caelus/run_caelus.sh` on Linux or Raspberry Pi, and `C:\Users\<name>\Caelus\run_caelus.cmd` on Windows.

The top of the dashboard shows the station name, the time of the last stored observation, and station status. **Station reporting** means Caelus is receiving usable observations. **Gateway standing by** means no current reading is available yet.

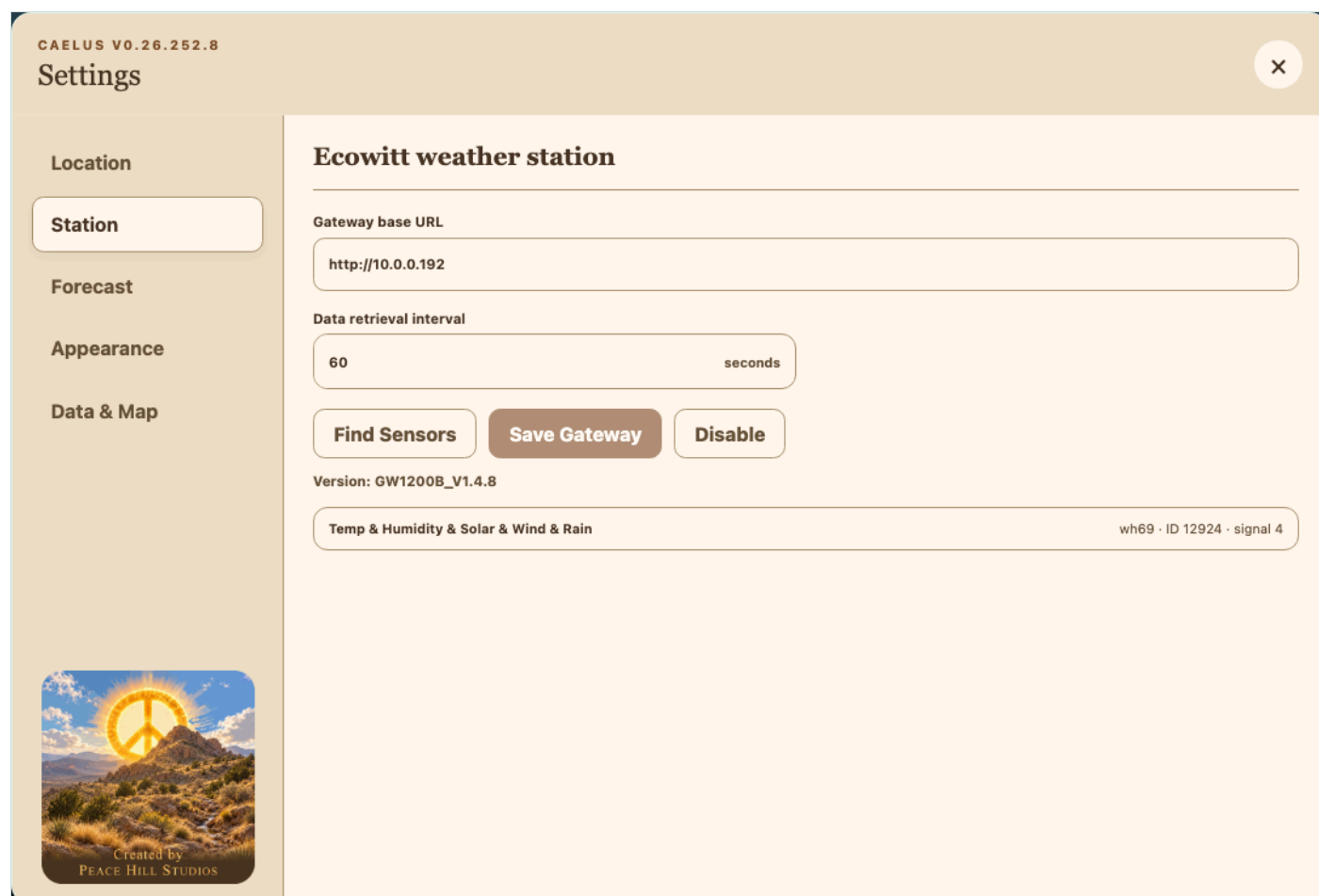
Quickstart Station Setup

Before configuring Caelus, use the Ecowitt app or gateway interface to:

1. Join the gateway to 2.4 GHz Wi-Fi.
2. Pair the outdoor weather array.
3. Confirm that the gateway displays live readings.
4. Give the gateway a stable LAN address, preferably with a DHCP reservation in the router.

Then configure Caelus:

1. Select **Settings**, then **Station**.
2. Enter the gateway base URL, such as `http://192.168.1.100`.
3. Select **Find Sensors**.
4. Review the reported gateway version and sensor inventory.
5. Set the retrieval interval from 60 to 3600 seconds.
6. Select **Save Gateway**.



The screenshot shows the Caelus V0.26.252.8 Settings interface. On the left is a sidebar with navigation options: Location, Station (selected), Forecast, Appearance, and Data & Map. The main content area is titled 'Ecowitt weather station'. It contains a 'Gateway base URL' field with the value 'http://10.0.0.192'. Below this is a 'Data retrieval interval' field set to '60 seconds'. At the bottom of the main area are three buttons: 'Find Sensors', 'Save Gateway' (highlighted in brown), and 'Disable'. Below the buttons, the version 'GW1200B_V1.4.8' is displayed. At the very bottom, a sensor list shows 'Temp & Humidity & Solar & Wind & Rain' with the identifier 'wh69 · ID 12924 · signal 4'. A small logo for 'PEACE HILL STUDIOS' is in the bottom left corner of the interface.

Station settings

Caelus uses read-only Ecowitt LAN endpoints. It does not configure Ecowitt cloud upload, custom-server push, MQTT, Nodus sensors, or switches. **Disable** stops future gateway polling without deleting stored history.

Settings

Select the **Settings gear** to the right of the Caelus title to open Settings. The Location pane opens first, and the dialog header shows the Caelus version. Each pane saves independently. A save in one pane does not submit unsaved edits from another pane.

The dialog supports mouse, touch, and keyboard navigation. Closing without saving restores an unsaved theme preview. Saved settings refresh the dashboard automatically while keeping the current pane open and preserving unsaved edits in other panes. Action and warning messages appear as toasts at the top center of the dialog and disappear after five seconds.

Station

The screenshot shows the 'Settings' dialog for Caelus V0.26.252.8. The 'Station' pane is selected in the left sidebar. The main area is titled 'Ecowitt weather station'. It contains a 'Gateway base URL' field with the value 'http://10.0.0.192'. Below it is a 'Data retrieval interval' field set to '60 seconds'. At the bottom of the main area, there are three buttons: 'Find Sensors', 'Save Gateway', and 'Disable'. Below these buttons, the version 'GW1200B_V1.4.8' is displayed. At the very bottom, a sensor status bar shows 'Temp & Humidity & Solar & Wind & Rain' and 'wh69 · ID 12924 · signal 4'. The left sidebar has a 'Location' section with 'Station' selected, and a 'Forecast', 'Appearance', and 'Data & Map' section. At the bottom left of the sidebar is a logo for 'PEACE HILL STUDIOS' featuring a peace sign over a landscape.

Station settings

- **Gateway base URL** must be an absolute `http://` or `https://` address without embedded credentials or a fragment.
- **Data retrieval interval** accepts 60 to 3600 seconds.

- **Find Sensors** queries the gateway and displays its model and paired inventory.
- **Save Gateway** saves a successful discovery and enables polling.
- **Disable** stops polling while preserving the database and saved history.

Both scheduled and manual polls use the same normalization and persistence workflow. A failed fetch does not erase the last stored observation.

Location

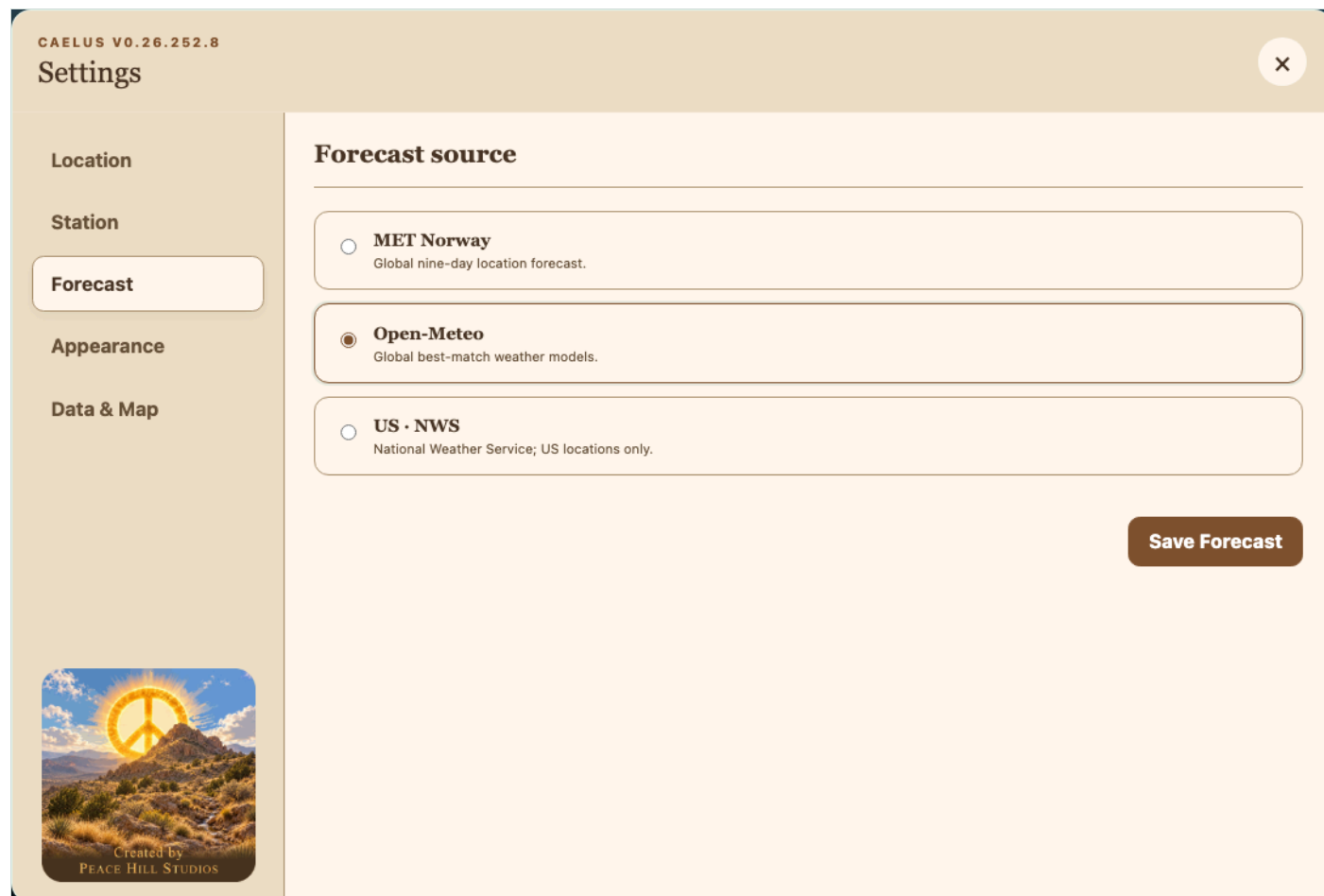
Location settings

- **Location name** is the dashboard heading.
- **Latitude** accepts -90 through 90.
- **Longitude** accepts -180 through 180.
- **Timezone** uses an IANA name such as *America/Denver*.
- **Use IP location** allows automatic detection.
- **Detect & fill from IP** performs detection immediately and fills the location fields.

For manual setup, clear **Use IP location**, enter coordinates and a matching timezone, then select **Save Location**.

When IP location is enabled or explicitly detected, Caelus sends the server's public IP address to `ipapi.co`, with `ipwho.is` as a fallback, to obtain an approximate city, timezone, latitude, and longitude. Astral then performs Sun and Moon calculations locally from the saved coordinates; Astral is not the IP-location provider.

Forecast



The screenshot shows the 'Settings' screen of the 'CAELUS V0.26.252.8' app. On the left is a sidebar with menu items: 'Location', 'Station', 'Forecast' (which is highlighted), 'Appearance', and 'Data & Map'. The main content area is titled 'Forecast source' and contains three radio button options: 'MET Norway' (Global nine-day location forecast), 'Open-Meteo' (Global best-match weather models, which is selected), and 'US · NWS' (National Weather Service; US locations only). A 'Save Forecast' button is located at the bottom right of the main area. At the bottom left of the screen is a small image of a peace sign over a desert landscape with the text 'Created by PEACE HILL STUDIOS'.

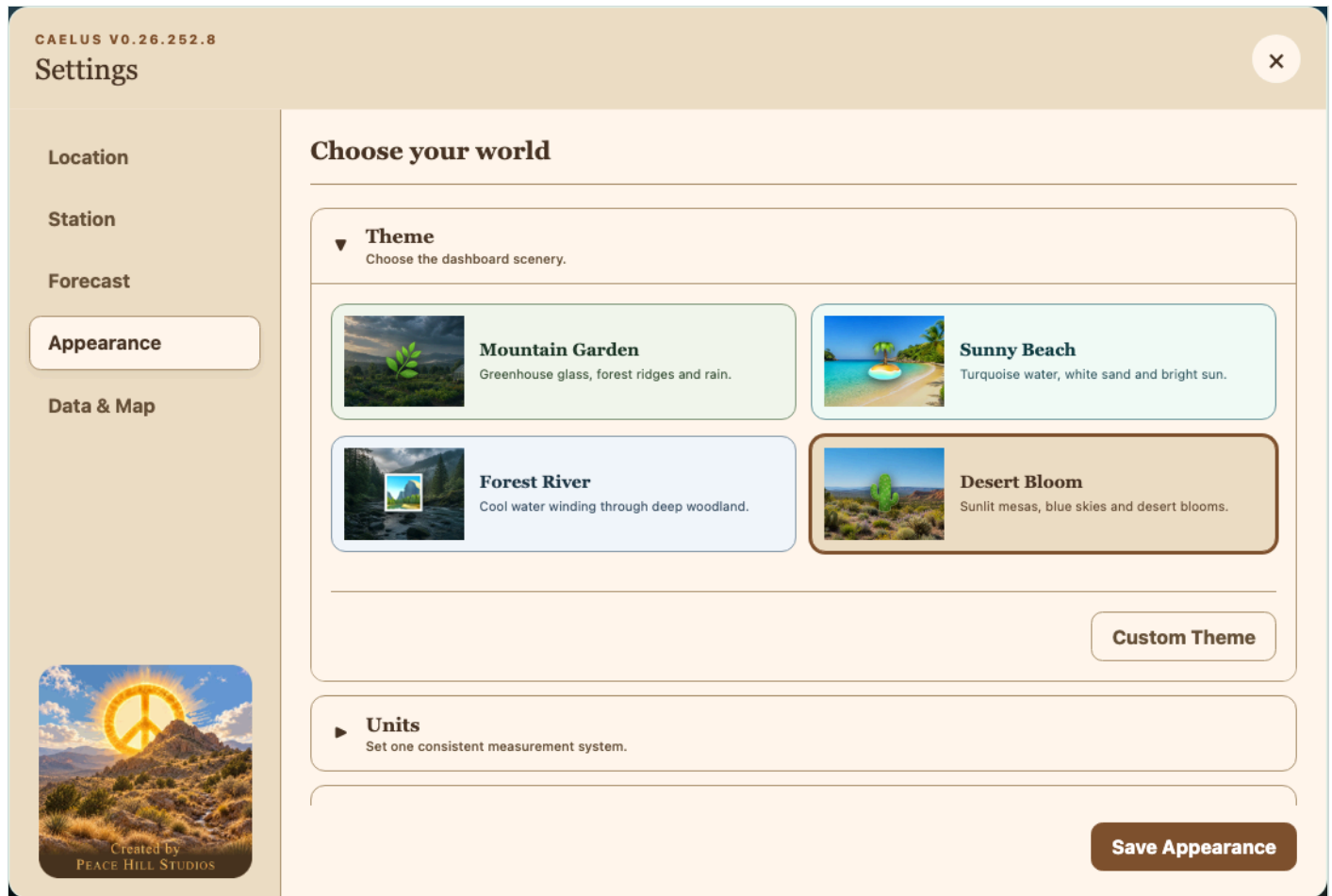
Forecast source settings

Choose one provider, then select **Save Forecast**:

- **MET Norway**: global location forecast.
- **Open-Meteo**: global best-match weather models.
- **US · NWS**: National Weather Service data for US locations only.

Forecast requests send the saved latitude and longitude to the selected provider. All provider responses are normalized into the same dashboard layout.

Appearance



Appearance settings with the Custom Theme action

The appearance pane contains three expandable sections:

- **Theme:** Mountain Garden, Sunny Beach, Forest River, Desert Bloom, or a custom image theme. Selecting any theme previews it immediately; **Save Appearance** makes it persistent.
- **Units:** Imperial (°F, mph, inHg, inches) or Metric (°C, km/h, hPa, millimetres).
- **Display Style:** set all metric cards to one initial style, then adjust individual exceptions.

Changing units, scenery, or card style does not modify the underlying weather history.

Create a Custom Theme

Select **Custom Theme** at the bottom of the expanded Theme section to open the creator.

CAELUS APPEARANCE

Create Custom Theme

Theme name

My Garden

Image selector

Choose File No file chosen

Image

Palette

Pale Sage

Image name

Image name

Remove Image

Add Image Create Theme

Create Custom Theme dialog

1. Enter a collection name, such as **My Garden**.
2. Choose an image, give it a display name, and select a palette. The palette controls the preview accents and is limited to Caelus's tested, readable choices.
3. Select **Add Image** to add more image choices to the same collection. A collection can contain one to five images, and each image can use a different palette.
4. Select **Create Theme**. Caelus validates and processes every image, then adds the collection to **Appearance > Theme**.
5. Select one of the new image cards and choose **Save Appearance** to make it the active dashboard theme.

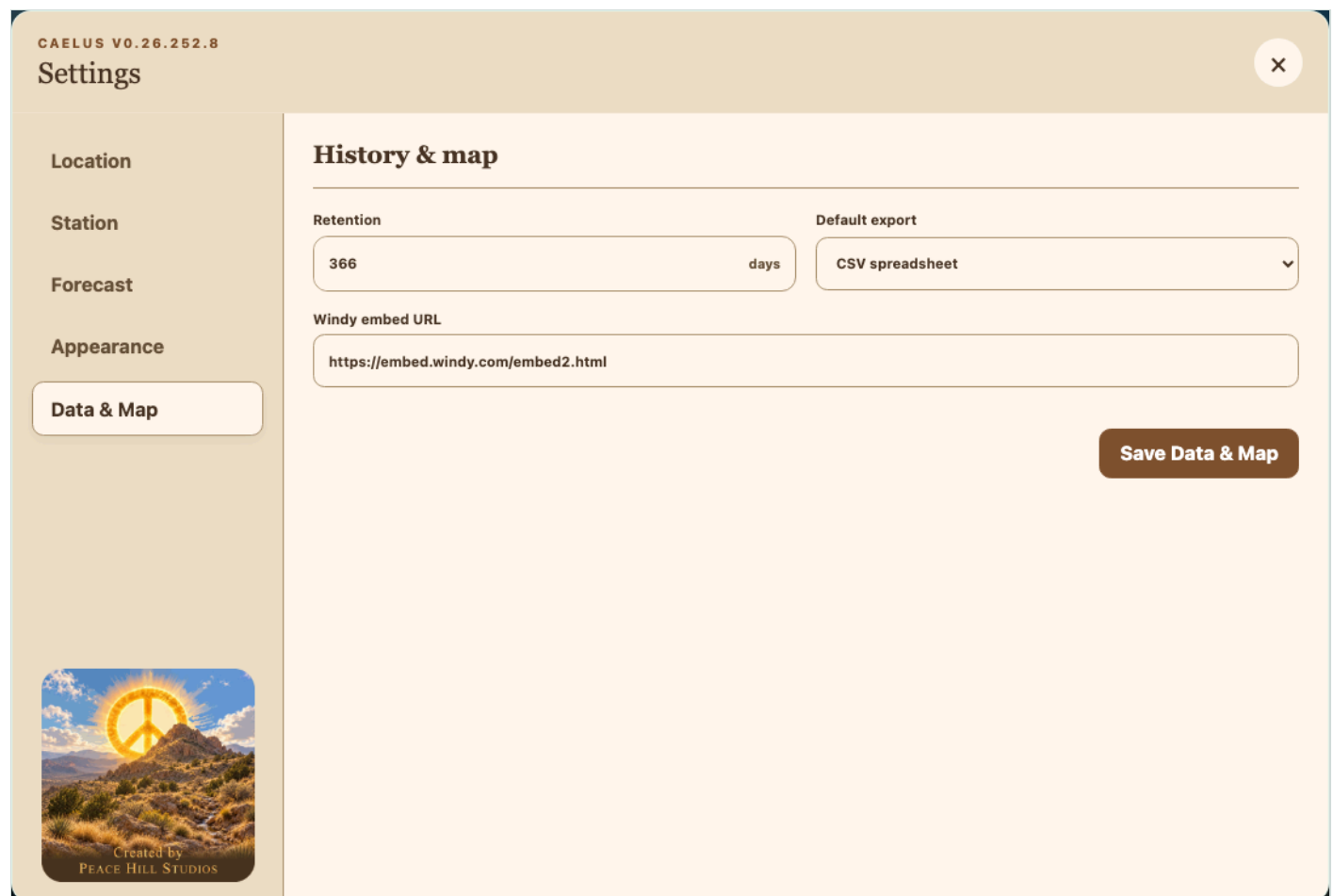
Custom Theme image requirements:

- WebP, JPEG, or PNG format.
- Static images only; animated images are rejected.
- At least 320 x 180 pixels and no more than 20 million pixels.
- No more than 5 MB per image.
- A 16:9 image at 1920 x 1080 is recommended. Other aspect ratios are center-cropped.
- Theme and image names are required and may contain up to 60 characters.
- Every image must use one of the eight supplied palettes: Pale Sage, Pale Earth, Pale Water, Pale Sky, Pale Blossom, Pale Fruit, Warm Neutral, or Cool Neutral.

Caelus converts accepted images to 1920 x 1080 WebP files and creates 480 x 270 WebP thumbnails. On a standard installation, the manifest is stored at /Users/<name>/Caelus/data/theme_settings/themes.json on macOS, /home/<name>/Caelus/data/theme_settings/themes.json on Linux or Raspberry Pi, or C:\Users\<name>\Caelus\data\theme_settings\themes.json on Windows. Processed images are stored beside it under the absolute data/theme_assets path for that installation. Back up the entire installed data directory to preserve custom themes with settings and history.

Select **Delete** beside a collection to remove that collection and its generated images. If the deleted collection is active, Caelus immediately falls back to Mountain Garden and saves that fallback. Deletion cannot be undone except by restoring a backup or recreating the theme.

Data & Map



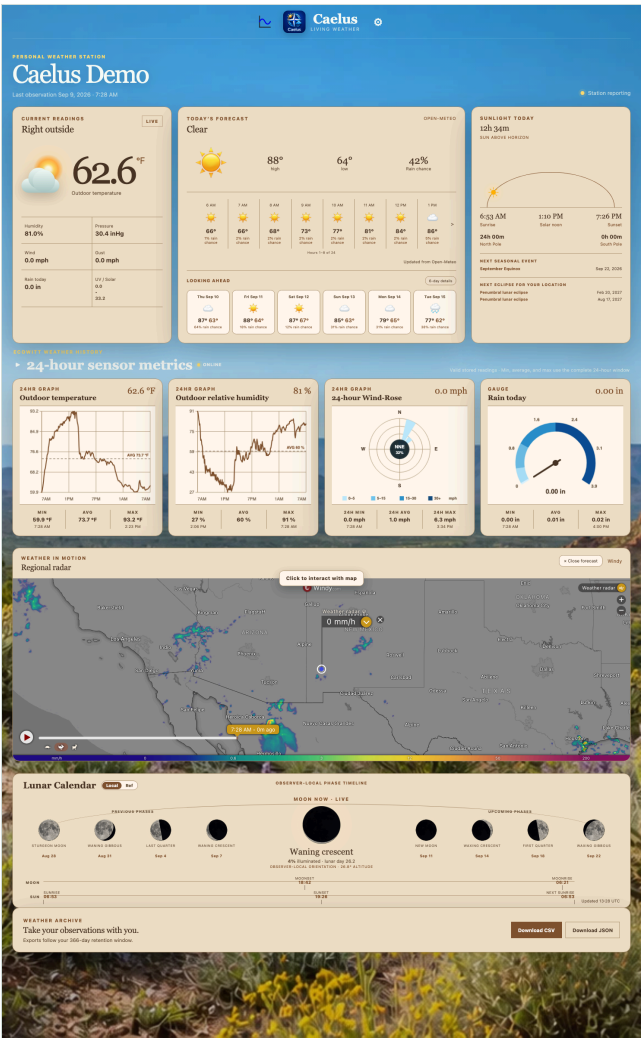
Data and map settings

- **Retention** accepts 30 to 366 elapsed days. Successful gateway polls prune readings older than this window.
- **Default export** selects CSV or JSON as the stored preference.
- **Windy embed URL** must remain the supported secure endpoint: <https://embed.windy.com/embed2.html>. Caelus adds the saved station coordinates and marker when it builds the displayed URL.

Select **Save Data & Map** to apply the pane.

Dashboard Overview

Current Readings



Full Caelus dashboard with the sensor row collapsed

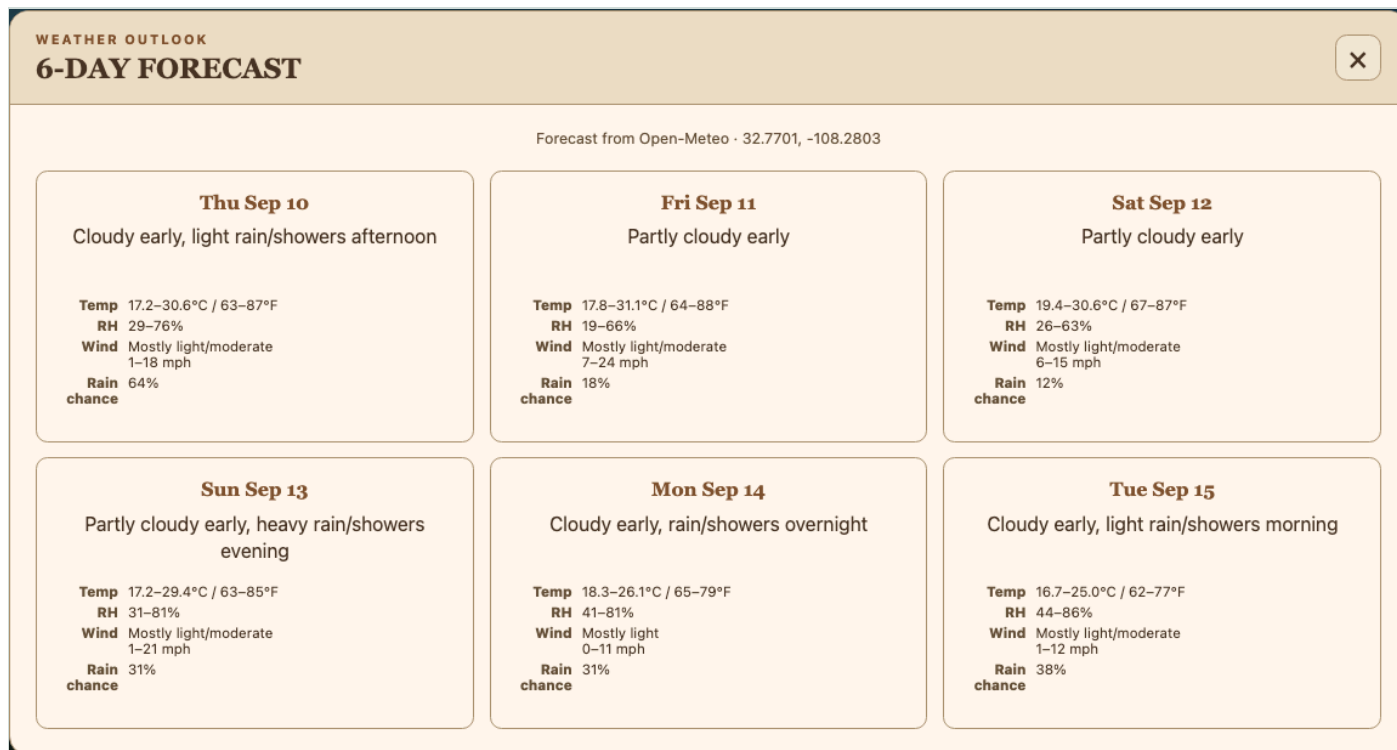
The **Current readings** card shows the latest outdoor temperature and a compact summary of outdoor humidity, relative pressure, wind, gust, rain today, UV, and solar radiation. A dash means the gateway did not supply a usable value for that field.

Values are displayed in the unit system selected under **Settings > Appearance > Units**. Changing units changes presentation only; stored readings are not rewritten.

Today's Forecast

The forecast card shows today's condition, high and low temperature, precipitation chance, up to 24 hourly samples, and six future daily summaries. Use the arrow at the right of the hourly row to page through later hours.

Select **6-day details** for daily temperature ranges, relative-humidity ranges, wind descriptions and speeds, precipitation chances, and early/late sky summaries.



Six-day forecast details

The provider label identifies the active forecast source. When a provider is temporarily unavailable, Caelus preserves and labels the last good cached forecast rather than replacing it with an empty result.

Sunlight and Seasonal Information

The **Sunlight today** card uses the saved coordinates and timezone. Sunrise, solar noon, sunset, daylight duration, and the daylight track come from the same local-date calculation. It also shows daylight at the North and South Poles, the next seasonal event, and locally visible eclipse information when the astronomy data package is available.

If eclipse calculations are unavailable, rerun the Caelus installer so the packaged astronomy data is installed in the Caelus runtime.

24-Hour Sensor Metrics

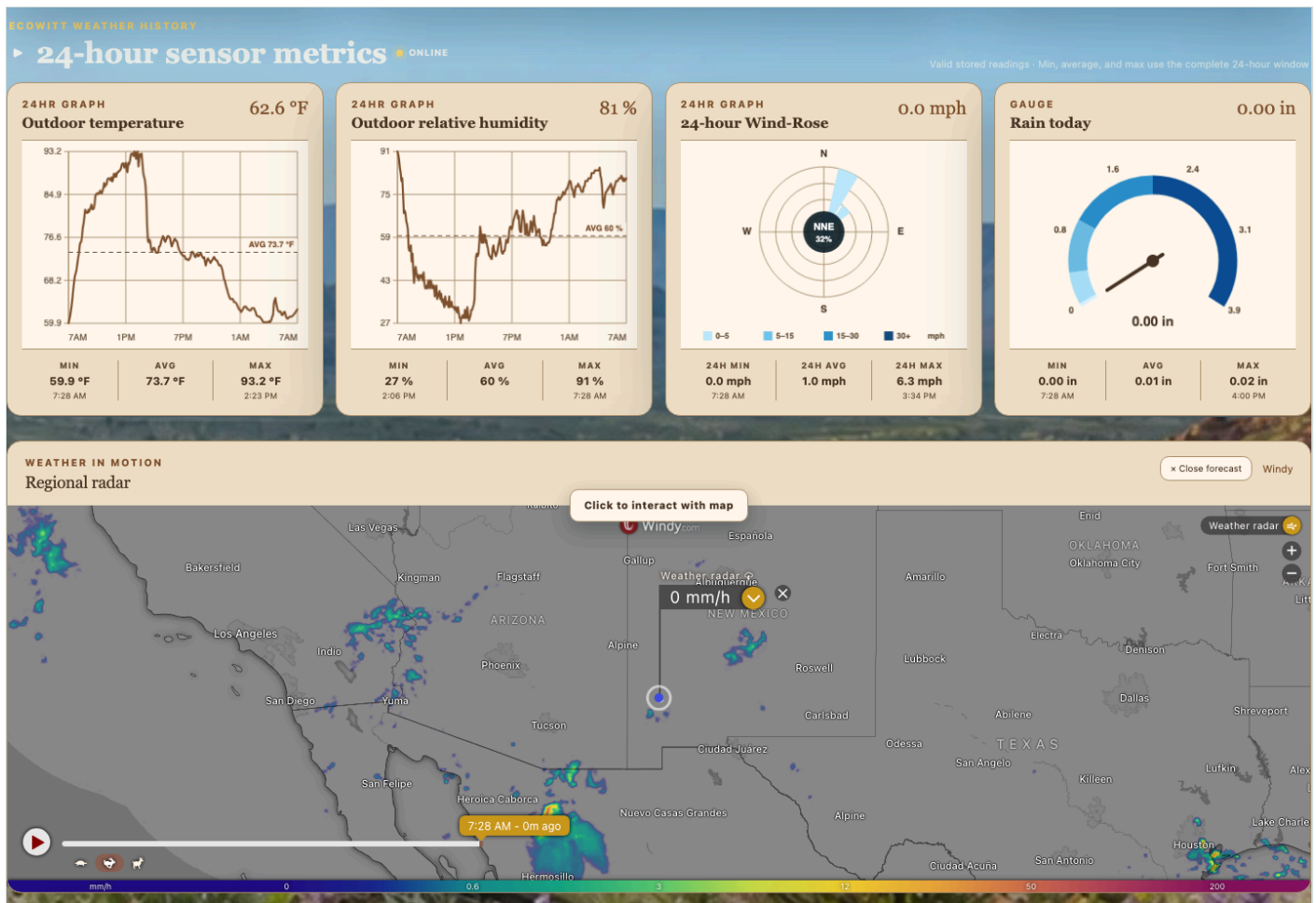
The **24-hour sensor metrics** section shows every metric that has valid stored data. The first four cards are visible initially. Select the triangle beside the section title to reveal or hide the remaining metrics.

Each card includes the current value and the minimum, average, and maximum calculated from the complete 24-hour window. Minimum and maximum timestamps use the configured local timezone.

A card can display:

- **24Hr Graph**: stored readings for the last 24 hours.
- **6Hr Graph**: stored readings for the last six hours.
- **Gauge**: the current value on that metric's display scale.

Select a metric card to cycle its current view. To make the starting style persistent, use **Settings > Appearance > Display Style**.



Metric cards and regional radar

Regional Radar

The Windy map is centered on the saved station coordinates and starts with its interaction guard enabled so normal page scrolling is not captured accidentally.

1. Select **Click to interact with map** before panning, zooming, or opening a Windy forecast.
2. Move the pointer away from the map or press Escape to restore the scroll guard.
3. Select **Close forecast** above the map to close an open Windy forecast panel and return to the station-centered radar view.

Windy is a separate cross-origin service. Caelus cannot directly operate the controls inside its forecast panel.

Moon Phase and Weather Archive

The **Lunar Calendar** header shows four phases before and four phases after the live Moon. The center disk is calculated for the saved observer location: illumination, lunar age, altitude, bright-limb direction, and lunar-surface rotation reflect the local sky instead of a fixed phase icon. Choose **Local** for that observer-local view or **Ref** for a

conventional upright phase diagram. The Sun and Moon rows share a sunrise-to-next-sunrise scale, placing sunset, moonrise, and moonset in chronological context.



Observer-local Moon and archive controls

Use **Download CSV** or **Download JSON** to export observations inside the configured retention window. CSV is intended for spreadsheets; JSON preserves a machine-readable object structure. Browser downloads use the button selected, even if the default export setting differs.

Caelus Graphum

Select the **Graph glyph** to the left of the Caelus title to open the full-screen history graph.



Full-screen weather history graph

Horizontal grid lines appear when a single metric is graphed. With multiple metrics, those horizontal lines are omitted. Every station-local day boundary has a vertical line and a date tick in ddmmm format, such as 09Sep.

Choose a time window of 1, 6, 12, or 24 hours, or 3, 7, 14, or 29 days. Select up to four metrics. Caelus gives each selected measurement an independently labeled scale so values with different units can be compared over the same time period.

The available list contains only metric types Caelus knows how to normalize; a selected metric can still have gaps when the gateway did not report it. Close the graph with the **X** button or Escape.

Mobile Operation

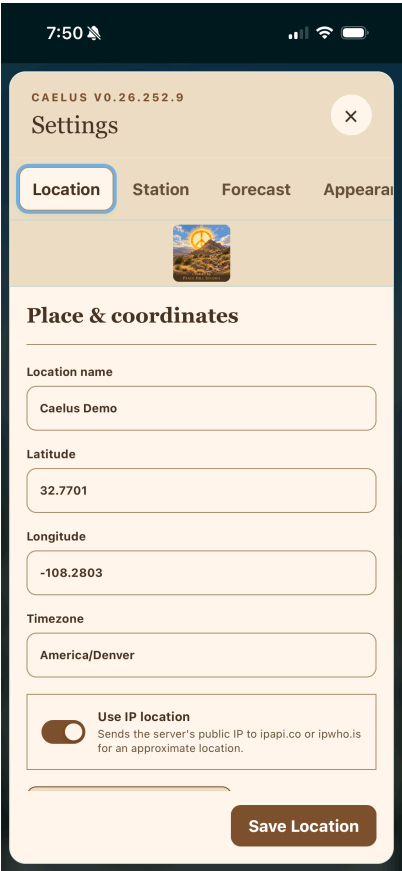
Mobile access is available only while your phone is connected to the same local area network (LAN) as the Caelus host. Cellular data or Wi-Fi on another network will not provide access. Adding Caelus to the Home Screen does not make it available remotely or offline.

Connect the phone to that LAN's Wi-Fi, then open Caelus in its browser or from the Home Screen icon. Leave Caelus running on the host. Use the host's network address, such as `http://samhain.local:8767`; `127.0.0.1` on the phone refers to the phone itself.

Mobile dashboard



Mobile settings



Dashboard and Refresh

- **Scroll vertically** to move through current readings, forecasts, sensor metrics, the map, and the lunar calendar. The dashboard stacks its cards to fit a narrow screen.
- **Tap the Caelus icon or title** to reload the dashboard. Use this to pick up a location, forecast source, or other setting changed on another device, without closing or restarting the web app. Readings also refresh automatically while the dashboard is active.
- **Tap the Graph glyph** to the left of the title to open Caelus Graphum. Choose a time range and up to four metrics; tap **X** to return to the dashboard.
- **Tap a sensor card** to cycle its graph or gauge view. Use the triangle beside **24-hour sensor metrics** to show or hide additional cards.
- **Tap Click to interact with map** before panning or zooming Windy. Use **Close forecast** above the map to reset an open Windy forecast panel.

Mobile Settings

1. Tap the **Settings gear** to the right of the title. Settings opens on **Location**, with the Caelus version shown at the top.
2. Swipe the **menu row horizontally** to reach Station, Forecast, Appearance, and Data & Map. Tap a menu choice to open its pane.
3. Scroll **within the pane** to reach its fields and controls. Edit the values, then tap that pane's **Save** button. Each pane saves independently; Station uses **Find Sensors**, then **Save Gateway**.
4. Look for the confirmation or warning toast at the top center of the dialog. Messages disappear after five seconds. A successful save refreshes the dashboard and keeps the current pane open.
5. Tap the **X** at the top right to close Settings. Closing does not save pending edits, and an unsaved theme preview returns to the saved theme.

Settings belong to the Caelus host and are shared by its connected devices. After saving on the Mac or another phone, tap the **Caelus icon or title on each already-open dashboard** to display those changes there. Use this refresh control in the Home Screen web app rather than relying on a browser pull-down gesture.

Routine Operation

- Leave Caelus running on the host computer so scheduled polling and retention continue.
- Give the gateway and Caelus computer stable LAN addresses when possible.
- Check the station-status label and last-observation time before relying on a reading.
- Treat forecast data as guidance; it is supplied by the selected external provider, not by the Ecowitt gateway.
- Back up the installed data directory if the local history matters. Typical paths are `/Users/<name>/Caelus/data` on macOS, `/home/<name>/Caelus/data` on Linux or Raspberry Pi, and `C:\Users\<name>\Caelus\data` on Windows.

Troubleshooting

The Dashboard Does Not Open

Confirm that the Caelus launcher is still running and open `http://127.0.0.1:8767` on the host. From another device, use the Caelus computer's LAN address and confirm that both devices are on the same trusted network. A host firewall may need to allow inbound TCP port 8767.

If Caelus was installed elsewhere, use the absolute launcher path for that installation rather than a relative command.

The Station Is Standing By or the Observation Is Old

1. Confirm that the Ecowitt gateway is powered and shows sensor data in its own interface.
2. Confirm that the gateway and Caelus computer are on the same LAN.
3. Open **Settings > Station** and verify the gateway's current base URL.
4. Select **Find Sensors** and review the reported model and inventory.
5. Select **Save Gateway** after a successful discovery.

If the gateway received a new DHCP address, update the URL and create a DHCP reservation to prevent another change.

Forecast, Sun, Moon, or Map Information Is Wrong

Open **Settings > Location** and verify the latitude, longitude, and timezone together. Then verify that the chosen forecast provider supports the location. The US National Weather Service option is limited to US coordinates.

A Metric Is Missing

Caelus displays a metric only after the gateway supplies a valid value and Caelus stores it. Expand the metric section, confirm the sensor reports that measurement in the Ecowitt interface, and allow at least one successful poll. Malformed or unavailable gateway fields are skipped without stopping the poller.

Health Check

Open `http://127.0.0.1:8767/healthz` on the Caelus computer. A healthy running poller returns HTTP 200 with `{"status":"ok"}`. A stopped, failed, or cancelled poller returns HTTP 503 with a diagnostic detail.

Privacy and Network Safety

Caelus is intended for a trusted private LAN and does not provide its own login screen. Do not expose port 8767 directly to the public internet.

External requests can disclose:

- The server's public IP address to the location provider when IP location is used.
- Saved latitude and longitude to the selected forecast provider.
- Saved latitude and longitude to the embedded Windy map.

Use manual coordinates and disable IP location if you do not want to use public IP-geolocation services.